Reviewer Pack — Minimal Order of Automorphic Lattices Bounds Resolution Proo…

Entry 565f566b2157 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

2026-05-29 20:36:16 UTC

Minimal Order of Automorphic Lattices Bounds Resolution Proof Width

Entry ID: `565f566b2157` **Verdict**: `INCONCLUSIVE` **Recorded**: 2026-05-29 20:36:16 UTC

1. Conjecture

Field A (mathematical branch): Automorphic Lattice Theory **Field B** (complexity object): Boolean Function Complexity: Tree-like Resolution Proof Complexity

Statement:

> For a given Boolean function φ, the minimum order of an automorphic lattice that contains φ as a sub-lattice is upper-bounded by the resolution proof width of φ, i.e., O(1) ≤ MinOrderLat(φ) ≤ t*(φ).

Rationale (proposer's reasoning):

> Automorphic lattices provide a geometric framework for understanding the structure of Boolean functions. A deeper connection between their order and the complexity of refuting these functions could shed light on the intrinsic difficulty of computational problems in complexity theory.

Taxonomy category: `cg_kw_andreev` (status at proposal time:)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): `78b160534c976eba`

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

> The conjecture is supported if, for all generated Boolean functions φ with n variables, the correlation between MinOrderLat(φ) and t*(φ) across 30 seeds is statistically significant (p-value ≤ 0.05), with an average MinOrderLat(φ) at most O(t*(φ)) within 3 standard deviations.*

3. Barrier filter (F1)

Final: `PASS`

Barrier	Verdict	Confidence	LLM₁	LLM₂
RELATIVIZATION	SAFE	0.50	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.50	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	UNCERTAIN	0.80	HITS	UNCERTAIN
KARP_LIPTON	SAFE	0.50	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: `ADJACENT_OK` against 9 hits across arXiv + Semantic Scholar + ECCC

Search queries (3): - `automorphic lattices AND resolution proof width` - `Boolean function complexity AND automorphic lattice theory` - `tree-like resolution AND minimal order of lattices`

Top relevant hits considered: - [http://arxiv.org/abs/1604.04253v2] On Period Relations for Automorphic L-functions II - [http://arxiv.org/abs/1212.6525v1] Automorphic Integral Transforms for Classical Groups I: Endoscopy Correspondences - [http://arxiv.org/abs/1511.00586v1] Strong local-global phenomena for Galois and automorphic representations - [http://arxiv.org/abs/2003.09703v1] Variance function of boolean additive convolution - [http://arxiv.org/abs/1407.6169v3] Multiplicative Complexity of Vector Valued Boolean Functions - [http://arxiv.org/abs/1210.6969v2] Fisher zeros and conformality in lattice models - [http://arxiv.org/abs/2204.05464v2] Lipschitz Functions on Quasiconformal Trees - [http://arxiv.org/abs/2407.17947v2] Supercritical Size-Width Tree-Like Resolution Trade-Offs for Graph Isomorphism

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)
Sandbox: isolated subprocess, stdlib-only, ≤90s wall time per cycle **Execution**: rc=1, elapsed=0.1s

5.1 Generated Python source

```python
import random
import math

def run_trial(seed: int) -> dict:
    random.seed(seed)

    def generate_boolean_function(n):
        return [random.choice([0, 1]) for _ in range(2**n)]

    def resolution_width(phi):
        n = int(math.log2(len(phi)))
        clauses = []
        for i in range(n):
            clause = [phi[j] if j < (i + 1) * n else not phi[j] for j in range(2**n)]
            clauses.append(clause)
        return len(clauses)

    def min_order_lat(phi):
        n = int(math.log2(len(phi)))
        # Simple heuristic: order is proportional to the number of variables
        return n

    phi = generate_boolean_function(random.randint(5, 10))
    t_phi = resolution_width(phi)
```

```
    MinOrderLat_phi = min_order_lat(phi)

    return {
        "metric_name": "MinOrderLat",
        "metric_value": MinOrderLat_phi,
        "instances_tested": 1,
        "n_max": n,
        "conjecture_holds": MinOrderLat_phi <= t_phi,
        "counterexample": "" if MinOrderLat_phi <= t_phi else f"phi={phi}, t*(phi)={t_phi}, MinOrd
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i + 1 for i in range(5, 8)]

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {result}")
        results.append(result)

    if all(r["conjecture_holds"] for r in results):
        mean_value = sum(r["metric_value"] for r in results) / len(results)
        std_value = math.sqrt(sum((r["metric_value"] - mean_value)**2 for r in results) / len(resu
        support_fraction = 1.0
    else:
        mean_value = None
        std_value = None
        support_fraction = sum(r["conjecture_holds"] for r in results) / len(results)
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)

    print(f"RESULT: {'SUPPORTED' if all(r['conjecture_holds'] for r in results) else 'FALSIFIED'}
```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

```
--STDERR--
Traceback (most recent call last):
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_15fd7a86.py", line 56, in <module>
    result = run_trial(seed)
             ^^^^^^^^^^^^^^^
  File "/home/ludo/Scrivania/SEC/research/pvsnp_sandbox/test_15fd7a86.py", line 45, in run_trial
    "n_max": n,
             ^
NameError: name 'n' is not defined
```

8. Critic adversarial review

Critic verdict: `CHALLENGE`

Critic reasoning:

> skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: `INCONCLUSIVE`

Reasoning:

> The test code crashed before producing data, which prevents us from evaluating the statistical significance of the correlation between MinOrderLat(φ) and t*(φ). As a result, we cannot confirm or refute the conjecture based on the provided evidence. | next: Re-run the test with proper error handling to ensure that it completes without crashing. Once the test is successful, re-evaluate the statistical significance of the correlation.

11. Audit log (LLM calls)

Total LLM calls: 9

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	glm4:latest	0	0	20435
2	preregistration	ollama_remote	glm4:latest	0	0	9956
3	novelty	ollama_remote	glm4:latest	0	0	14395
4	novelty	ollama_remote	glm4:latest	0	0	13570
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	18363
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	14241
7	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	6777
8	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	12765
9	judge	ollama_remote	glm4:latest	0	0	9555

Totals: 0 input tokens, 0 output tokens, ~$0.0000 cost, 120055 ms total latency. Provider mix: {‘ollama_remote’: 9}

(full prompt+response transcripts available in `research/audit/565f566b2157.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/565f566b2157.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)
Replay tarball: `research/replay/565f566b2157.tar.gz` (if generated)